Wildfire smoke: interpretation of current literature and statistical modeling of Canadian interprovincial data

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Abstract

Climate change exacerbates extreme weather events and has led to significant increases in international forest fire rates. The period 2021–2024 marked a record high for the amount of land burned in Canada due to forest fires [1]. This report seeks to examine the economic and health-related stress of forest fires as potentially intensifying factors of substance abuse and opioid-related mortality.

After investigation of topical literature, this report transitions to quantitative modeling of aggregate Canadian interprovincial data 2019-2024. This statistical modeling experiment was performed to investigate the potential correlative relationship between lagged forest fire rates, mental health distress (quantified by acute mental-health hospitalizations) and annual opioid-related death rates in Canada. Forest fire rates were lagged by 1 year to create temporal precedence based on the hypothesis that stress caused by forest fires would negatively influence mental health and substance abuse after a certain period of time, and by extension, increase substance-related mortality. Modeling was performed for all Canadian provinces excluding Yukon and Northwest Territories, due to a lack of publicly accessible forest fire data. Mediation analysis was employed to measure associations and demonstrate a potential mechanism of change across a variety of pathways (**Figure 1**). Data produced through 20,000 iterations of non-parametric bootstrapping simulation determined that this mental health proxy pathway demonstrated an ≈13.7% proportion mediated.

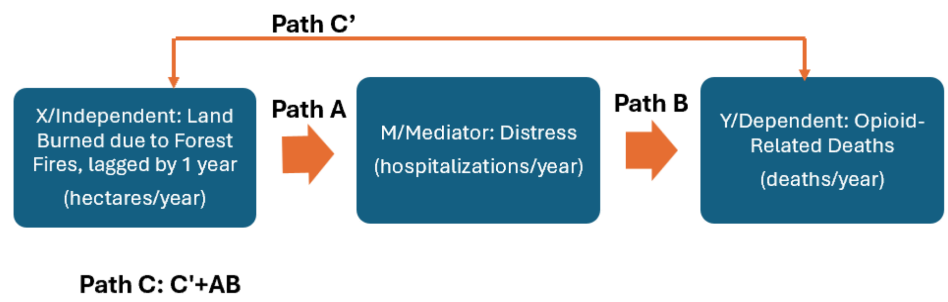

Figure 1-Mediation Analysis Pathways: (A x B)=**indirect effect** of X on Y, Path C'=**direct effect** of X on Y, Path C=**total effect** of X on Y.

1) Introduction

1.1) Climate change, wildfires, and economics

Economic decisions on behalf of individuals and governments have worsened the impacts of climate change and natural disasters. For instance, uncontrolled burning of rainforests and unsustainable farming practices make forest fires more difficult to control. Peat, 2-3m deep topsoil created by unsustainable farming practices, is flammable, thereby making fires more difficult to extinguish. Additionally, warming weather and changes in wind patterns create high-pressure regions that prevent smoke dissipation. Smoke in urban areas traps car and factory emissions, thereby catalyzing their consequences on health [5].

Computational modeling provides valuable insight into the economic devastation of forest fires. The US Social Cost of Carbon framework (USCC) approximates economic damage resulting from 1 metric ton of $Co2$ emissions. Computational modeling of integrating fine particulate matter (PM2.5) into the USCC demonstrated a 74% increase in monetary value [7]. Moreover, it is estimated that for every metric ton of $Co2$ released in 2025 caused ≈$11.20 worth of damage specifically through wildfire smoke mortality, as quantified by the value of a statistical life defined by the Environmental Protection Agency (EPA) [8]. Additionally, modeling of national PM2.5 impacts from wildfire smoke in Canada from 2013–2015 estimated health impacts to be $410M–1.8B for acute health impacts and $4.3B-$19B for chronic health impacts [11].

Recent years have had particularly devastating consequences on both the national and regional economies of Canada due to forced evacuation, loss of housing, and loss of working days. It is estimated that, in 2023, ≈232,000 citizens nationwide had to evacuate their homes with ≈$945 million in insured damages in British Colombia, the Northwest Territories, Okanagan, Shuswap, Behchoko-Yellow Knife, Tantallon, and Hammonds Plains. Economic loss was especially severe in the Northwest Territories, where ≈74% of total economic activity was in wildfire-affected areas. In Yellowknife and Kelowna in 2023, as well as Jasper in 2024, >6% of working days were lost on an annual basis due to evacuations. Similarly, by June 2025, ≈25% of Manitoba and northern Saskatchewan regional economies were at risk due to wildfires [9].

1.2) Wildfire smoke and physical health

Continued increase in global temperatures is likely to contribute to worsening wildfire smoke-related mortality. According to the National Bureau of Economic Research (NBER), there is predicted to be a roughly 76% increase in wildfire smoke-related deaths in the next 2–3 decades [6]. Furthermore, predictive modeling theorizes that a 3°C in global warming would cause ≈64,000 smoke-related deaths/yr and a 2°C increase in global warming would cause ≈14% less deaths/yr than the 3°C model (95 CI) [7]. As explained by N. Harris MD from Mass General Brigham, “We’re only as healthy as the environment around us…climate change is a healthcare emergency, wildfires are a symptom” [6].

Forest fire smoke is composed of air pollutants including nitrogen oxides, carbon monoxide, ozone, aromatic carbons, and lead [2], causing it to be ten times more toxic than pollution emitted from the burning of fossil fuels [3]. Additionally, fumes from burning plastics can get caught in smoke and make smoke even more toxic [6]. Forest fire smoke is by no means a localized issue as smoke can travel over 1,000 miles [6]; wildfire pollutants from Northern Canada in 2015 reached North and South Dakota, Minnesota, and Iowa [10].

Acute impacts of exposure to wildfire smoke include coughing, trouble breathing, asthma attacks, stinging eyes, chest pain, and fast heartbeat. This is emphasized in a 2022 investigation into short-term heart rate variability (HRV). Participant heart activity was monitored in both areas, with and without air pollution. The study demonstrated that individuals exposed to Australian bush fires experienced worse Standard Deviation of Average Normal-to-Normal Sinus Intervals (SDANN) than those in a similar area without smoke exposure [10]. Similarly, exposure to PM2.5 in Canada has been associated with increased rates of respiratory and cardiovascular mortality and morbidity [11]. Moreover, long-term exposure to forest fire smoke can include physical complications such as cancer, dementia, reduced immune system function, and heart, lung, and kidney disease [6].

The physical impact of forest fires pose the greatest risk to those with pre-existing conditions such as asthma, diabetes, pregnancy and chronic kidney or heart disease [4]. An analysis of the impacts of forest fires from the period 1967-1977 in Kuala Lumpur revealed that the day after a high-pollution episode, the mortality of individuals aged 65–74 increased 57% with deaths due to non-traumatic factors increasing 72% [5].

1.3) Climate change, mental health, and hypothesis formation

Changes in mental health influence human behavior. Forest fires and other natural disasters are associated with increases in mental illness such as mood disorders, anxiety, and depression [6]. Of particular risk for mental health complications during times of natural disaster are pregnant women. The stress of social cohesion, family support, and childcare during times of natural disaster have been proven to increase morbidity for pregnant women and their children. Women pregnant, who had given birth, or were within 3 months of conceiving during the 2019-2020 Australian bush fires reported 12% higher depression, 35% higher anxiety and 43% higher stress than Australian norms, according to DASS-21 and WHO-5 surveys [13].

Data analysis can be used to draw conclusions between climate change and mental health. For instance, a 2021 global meta-analysis of 1.7 million mental-health related mortality demonstrated that for every 1°C increase in temperature there is a 2.2% increase in mental-health related mortality (95%CI)[12]. Furthermore, global temperature was proved to be associated with an increase in diagnoses of mood disorders, organic mental disorders, such as Alzheimer's and Parkinson's, schizophrenia, and anxiety disorders. The greater mortality risk was associated with substance-related mental disorders; individuals with such disorders were associated with a 4.6% increase in mortality (95%CI) [12]. Further statistical modeling based on California death certificates from 2014 to 2019 demonstrated that a 5°C increase in temperatures was associated with a 3.1% increase in deaths due to suicide [15].

This modeling experiment was based on the hypothesis that an increase in forest fire rates would increase the stress of populations, thereby leading to increased associations with opioid-related mortality. This hypothesis has been previously tested using a Monte Carlo simulation in 2024 using information about teenagers in the United States, predicting a 12–34%

increase in opioid-related mortality after exposure to a wildfire [14]. However, unlike the previously mentioned study, which aimed to create predictive data, the bootstrapping methods used in this experiment seek to identify trends within aggregate existing data from Canadian records. In this way, it was intended to increase sample size and eliminate the uncertainty associated with predictive modeling. Forest fire data was acquired through Canadian Wildfire Information Systems (CWIS) and hospitalization data was collected from Canadian Institute for Health Information (CHI), and opioid-related mortality data was obtained from Public Health Agency of Canada (PHAC) coronary records. In using existing data in place of generating theoretical data, it was hoped that this modeling experiment could provide more accurate insights into the real impacts of natural disasters, specifically forest fires, on mental health and human behavior.

2) Experimental approach and algorithm components

2.1) Variables, scaling and lagging

Mediation analyses utilize an intermediate value (M) to connect the theoretical relationships between the independent (X) and dependent variable (Y). It was theorized that increased forest fire rates (X) would lead to increased distress (M), quantified through mental health-related hospitalization statistics, and this increase in distress would lead to greater rates of opioid-related deaths (Y), potentially reflecting increased misuse.

Forest fire rates were lagged by one year to create temporal precedence. Hospitalization rates and opioid-related death rates were standardized by estimated provincial populations as of July 1st of each year to create per-capita rates with a scale factor of 100k individuals.

2.2) Z-score standardization

Z-score standardization, or normalization, was employed to adjust for differences in scales, including rates of land burned, hospitalizations, and mortality, to ensure no one feature was given more importance than another. It was important that all features contribute equally to the results to avoid bias and make results less affected by outliers. This was done through calculation of mean, standard deviation, and then normalizing all values in the data set to create a mean of 0 and a standard deviation of 1 [16].

2.3) Models for point estimation: OLS regression and bootstrapping for indirect effect (AB)

Ordinary Least Squares (OLS) regression finds the line of best fit between coefficients through repeatedly minimizing squared residuals, according to the Gauss-Markov Theorem which explains OLS to be an unbiased, linear model [20]. OLS was run through 20,000 iterations of non-parametric bootstrapping, a type of randomized sampling with replacement, to calculate the point estimates [22]. In this experiment being non-parametric, used data was recycled back into the data set and given equal chance at being reselected during the following iteration. Bootstrapping was used to calculate the indirect effect of the independent variable, hectares burnt, on the dependent variable, opioid-related deaths. For each iteration, path A was run, path B and C' were run, and then path AB was calculated. This resampling enables the calculation of an empirical distribution function of the indirect effect; it is used to estimate the standard error of the mean [21]. Lastly, 95% confidence intervals (CIs) were calculated from the bootstrapping distribution to minimize outliers [18].

3) Results

3.1) Path C: Total effect of forest fire rates on opioid-related death rates (X→Y)

It was calculated that the total correlation from forest fire rates to opioid related death rates is 0.2980. A one standard deviation increase in forest fire rates is associated with a 0.298 standard deviation increase in opioid-related death rates.

3.2) Path C′: Direct effect of forest fire rates on opioid-related death rates

It was found that, after controlling for M, distress, for every one standard deviation increase in forest fire rates there is a 0.2571 standard deviation increase in opioid-related death rates.

3.3) Path AB: Indirect effect of forest fire rates on opioid-related death rates (X→M→Y)

Of the estimated total effect of 0.2980, it was determined that 0.0409 of that change is specifically explained by X on Y as mediated through M, a ≈13.7% proportion mediated.

4) Conclusions

4.1) Summarization of literature

Changes in climate and the wildfires associated with these changes have significant economic and social consequences. These changes impact human behavior as well as physical and mental health in a quantifiable manner and are predicted to continue to do so as global warming intensifies.

4.2) Limitations and statistical validity

Data proved statistically significant at the 95th confidence interval, excluding zero. Although the correlations identified are small in magnitude, they are statistically significant within the model. The direct effect, 0.257 was calculated to be ≈6.3x larger than the indirect effect, 0.049. The amount of data channeled through the theoretical pathway proved to be ≈13.7% leaving ≈86.3% of the data related to changes in the opioid-related mortality rate unexplained. This unexplained data is understandable as there are many factors such as economic factors, social factors, lack of needs being met by healthcare systems, and other environmental factors, not accounted for that influences changes in human mental health and behavior.

Limited publicly accessible data and computational power restricted this experimental modeling 20,000 iterations over 274,625 possible pathways. Future works could include expanding the sample size of the modeling. Also the mediator value, hospitalizations, only includes individuals in acute, not chronic mental distress. It could also prove useful to rerun the data with a different mediator such as the number of new anxiety diagnoses once the data for the referenced time frame is published.

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